

POWER ELECTRONICS

Switch Drivers

Pulse Width Modulation

- Lower the duty cycle, lower power delivered.
- High duty cycle means the load is on almost all the time
- Varying the duty cycle continuously changes the power delivered to the load
- This is called Pulse Width Modulation

H-Bridge

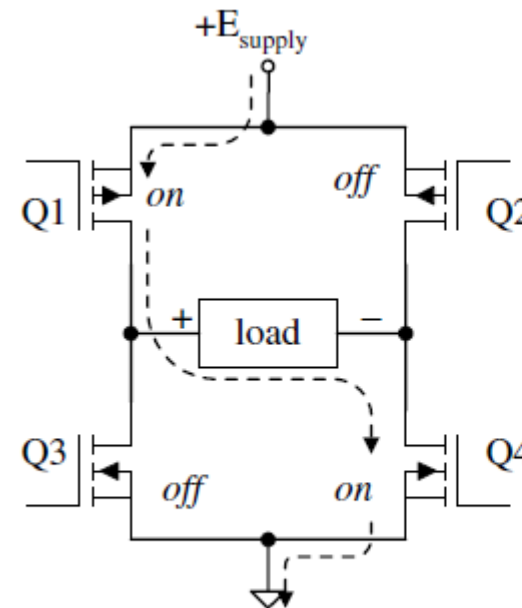
- High side switches and low side switches
- Start with all four transistors off
- To spin clockwise: Q1 and Q4 are on
- To spin counterclockwise: Q2 and Q3 are on
- What happens:

Q1 and Q3 are on at the same time?

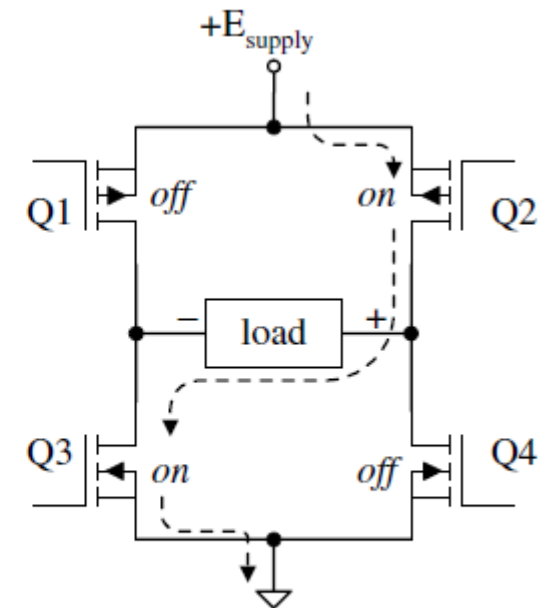
Or

Q2 and Q4 are on at the same time?

- Overall board is full of logic gates, timers, Drive transistors, load current sense circuit and Transistor temperature sense circuit



(a) clockwise



(b) counterclockwise

H-Bridge

- Allegro 3971 is a single IC that contains two independent H-bridges, made with n-channel MOSFETs
- Each transistor is protected with its own flyback diode
- Load supply is boosted to provide the gate drive for the high side transistors
- Load supply may be as high as 50 V dc
- Each bridge can output up to 2.5 A dc

Example

- The 3971 is to be used to drive a $15\ \Omega$ load from $\pm 28\text{ V}$ dc with a duty cycle that may vary up to 85%. Calculate the maximum power delivered to the load, dissipated by the 3971, and the heat sink thermal resistance needed. Assume a maximum ambient temperature of 60°C .
- $P_{IC\ max} = 2.2\text{ W}$, $T_{J\ max} = 150^\circ\text{C}$, $R_{on\ high\ side\ max} = 0.375\ \Omega$, $R_{on\ low\ side\ max} = 0.2\ \Omega$
- $\Theta_{JC} = 2^\circ\text{C/W}$ and $\Theta_{CS} = 0.2^\circ\text{C/W}$

Summary

- Diodes:
 - Too much reverse recovery time
 - Too much forward voltage drop
 - Not suitable in high speed switching
- Schottky Diode:
 - Drops only 0.3 V while carrying several amps
 - Switches in 10s to 100s of nanoseconds
- BJT
 - Base current produced by e_{in} and limited by R_{base}
 - β may be as low as 20 in high power transistors
 - Darlington pair is used to handle the low β issue
 - Switching speed $5\mu s$ or longer
 - Parasitic capacitance is major issue in low side BJT switch

Summary

- MOSFET:
 - Switching speed 50 ns even when handling 10 A
 - Parallel switches are more common than single unit with large heat sink
- H bridge:
 - Can send current in either direction with single supply
 - Can be built with discrete transistors, sensors, logic gates.
 - Allegro Microsystems offers fully integrated single and dual H bridge ICs.

Problem 1

- Design a low side switch using a 2N3904 to drive a $100\ \Omega$ load powered from a 5 V dc supply. Determine the collector current, base current, base resistor, power delivered to the load, and the transistor power dissipation for an 80% duty cycle pulse. $V_{CE(sat)} = 0.2\text{ V}$, $V_{BE} = 0.8\text{ V}$

Problem 2

- Design a low side switch using an IRF530 to drive a $35\ \Omega$ load powered from a 24 V dc supply. Determine the drain current, required gate pulse amplitude, power delivered to the load, and the transistor power dissipation for a 75% duty cycle pulse, and the required heat sink maximum thermal resistance for an ambient temperature of 60°C . ($R_{on\ at\ 60^\circ\text{C}} = 0.14\ \Omega$). $\Theta_{JC} = 62^\circ\text{C/W}$

Problem 3

- Design a switch using four IRF530s in parallel to drive a $35\ \Omega$ load from a 24 V dc supply with a 75% duty cycle. Calculate the voltage drop across the paralleled transistors, and the power *each* transistor must dissipate. Is a heat sink needed? Prove your answer with a calculation. $\Theta_{JC} = 62^\circ\text{C/W}$

Problem 4

- Design a low side switch using 2N3904 transistor to drive a load consisted of a resistor tied to 15 V dc supply and a 840 pF capacitor connected to common. Select the appropriate load resistor, and base resistor for a 2.4 V peak input, to provide the maximum rise time. Calculate that rise time.

For the 2N3904 switch

$$I_{C \max} = 100 \text{ mA}_p$$

$$V_{CE \text{ on}} = 0.3 V_p$$

$$V_{BE \text{ on}} = 0.9 V_p$$

Problem 5

- Design a *p*-channel high side switch to power a 5 V, 4 Ω load. For your design, assuming a 90% duty cycle to the load, calculate all components, rise time for gate voltage and, $R_{on @ T = 140^\circ\text{C}}$, P_{load} , P_Q , Θ_{SA} . ($C_{parasitic} = 860$ pF, Resistance is 1.84 times of that at 25 degree Celsius)

Problem 6

- Design a *pnp* high side switch to power a 5 V, 50 mA_p load. For your design, assuming an 80% duty cycle to the load, calculate all components, P_{load} , and P_Q .

- *SOLVE YOURSELF*

Reference

Power Electronics, Principles and Applications
J. Michael Jacob

Power Electronics
Ned Mohan